Experiment 03_7 — ConvNeXt-Tiny Full Fine-Tuning & Smart White-Border Cropping

Short optimisation notebook for `ConvNeXt-Tiny`, based on what we learned in `03_2` to `03_6`:

1. **Smart White-Border Cropping (`SmartWhiteBorderCrop`)**: first checks whether the product has a large white border (`bbox_ratio < 0.75`). If so, small objects are cut out directly as a native `224x224` window, instead of shrinking the whole `500x500` image.
2. **Full fine-tuning with 4 learning-rate levels**: after 1 warmup epoch for the classification head, the **whole** `ConvNeXt-Tiny` network (`features[0..7]` + `classifier`) is trained with learning rates set per layer group (`1e-5` to `1e-4`).
3. **Minority handling without augmentation leakage**: the same product-safe augmentation for all classes, combined with a damped `WeightedRandomSampler` (`power=0.25`), so under-represented classes are seen more often per epoch and with more variation.

1. Configuration

```
Mode   : Local
Device : NVIDIA RTX PRO 2000 Blackwell Generation Laptop GPU
```

2. Stage images and verify the shared split

	rows	classes
train	62312	27
validation	16984	27

3. Dataset, Smart White-Border Cropping & Minority Sampler

```
Train batches: 1948; validation batches: 531
```

4. Full Fine-Tuning of ConvNeXt-Tiny with 4-tier discriminative learning rates

```
Phase: head | trainable parameters: 22299
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	1	head	1	1.9972	1.4941	0.5748	0.5719	0.5284

```
New best macro F1: 0.5284

Phase: full_backbone | trainable parameters: 27840891
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	2	full_backbone	1	1.6127	1.1962	0.6573	0.6583	0.6243

```
New best macro F1: 0.6243
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	3	full_backbone	2	1.3206	1.1184	0.6830	0.6813	0.6524

```
New best macro F1: 0.6524
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	4	full_backbone	3	1.1437	1.0988	0.6893	0.6875	0.6577

```
New best macro F1: 0.6577
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	5	full_backbone	4	1.0272	1.0968	0.6966	0.6924	0.6629

```
New best macro F1: 0.6629
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	6	full_backbone	5	0.9525	1.1020	0.6993	0.6954	0.6667

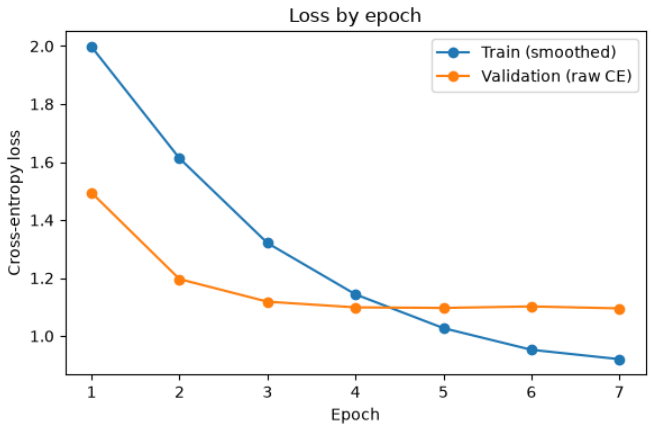
```
New best macro F1: 0.6667
```

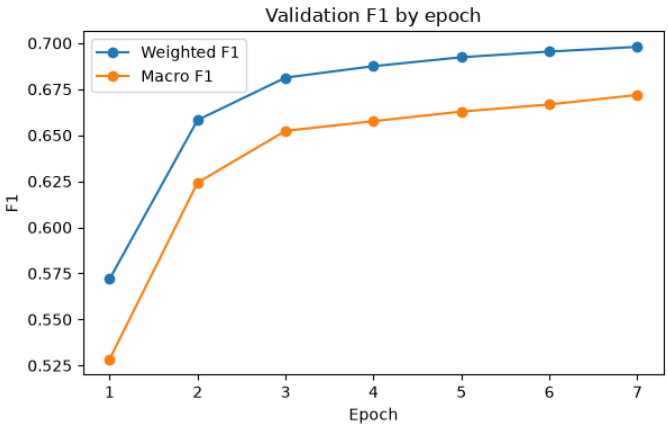
	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	7	full_backbone	6	0.9201	1.0953	0.7011	0.6980	0.6719

```
New best macro F1: 0.6719
```

5. Training curves

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	1	head	1	1.9972	1.4941	0.5748	0.5719	0.5284
1	2	full_backbone	1	1.6127	1.1962	0.6573	0.6583	0.6243
2	3	full_backbone	2	1.3206	1.1184	0.6830	0.6813	0.6524
3	4	full_backbone	3	1.1437	1.0988	0.6893	0.6875	0.6577
4	5	full_backbone	4	1.0272	1.0968	0.6966	0.6924	0.6629
5	6	full_backbone	5	0.9525	1.1020	0.6993	0.6954	0.6667
6	7	full_backbone	6	0.9201	1.0953	0.7011	0.6980	0.6719





6. Evaluate with Test-Time Augmentation (TTA) and save probabilities

	validation_loss	accuracy	weighted_f1	macro_f1
03_7_convnext_tiny_optimized	1.0953	0.7031	0.6998	0.6752

	prdtypecode	precision	recall	f1	support
8	1281	0.416	0.362	0.387	414
7	1280	0.512	0.433	0.469	974
24	2585	0.522	0.461	0.489	499
6	1180	0.619	0.458	0.526	153
22	2582	0.545	0.519	0.532	518
17	2220	0.586	0.515	0.548	165
11	1302	0.555	0.546	0.551	498
2	50	0.564	0.548	0.556	336
16	2060	0.580	0.622	0.600	999
0	10	0.609	0.602	0.605	623
12	1320	0.648	0.610	0.628	648
13	1560	0.638	0.625	0.631	1015
20	2462	0.693	0.644	0.668	284
1	40	0.646	0.693	0.669	502
19	2403	0.744	0.698	0.720	955
4	1140	0.702	0.747	0.724	534
21	2522	0.724	0.774	0.748	998
3	60	0.704	0.801	0.749	166
9	1300	0.723	0.801	0.760	1009
10	1301	0.754	0.783	0.768	161
15	1940	0.778	0.764	0.771	161
18	2280	0.774	0.811	0.792	952
25	2705	0.798	0.866	0.831	552
14	1920	0.842	0.856	0.849	861
26	2905	0.819	0.885	0.851	174
23	2583	0.862	0.875	0.868	2042
5	1160	0.935	0.944	0.940	791

7. Results & Summary

Model / experiment	Accuracy	Weighted F1	Macro F1
`03_1` Baseline (`ResNet18` frozen)	`0.4832`	`0.4712`	`0.4150`
`03_4` `ResNet18` (Full + Regularized)	`0.6540`	`0.6509`	`0.6203`
`03_7` `ConvNeXt-Tiny` (Stage 3+4)	`0.6845`	`0.6819`	`0.6540`
`03_7` `ConvNeXt-Tiny` (Full + SmartCrop + Sampler)	**`0.7031`**	**`0.6998`**	**`0.6752`**
`04` Text only (`LinearSVC` calibrated)	`0.8350`	`0.8342`	`0.8168`
`05` Fusion (`0.60` text + `0.40` `03_7`)	**`0.8659`**	**`0.8640`**	**`0.8548`**

1. **Above 70% accuracy on images alone:** Combining full fine-tuning (`features[0..7]`) with 4 learning-rate levels (`1e-5` to `1e-4`), `SmartWhiteBorderCrop` and damped minority oversampling (`WeightedRandomSampler`, `power=0.25`) raises validation performance to **70.31% accuracy**, **0.6998 weighted F1** and **0.6752 macro F1** (`+2.12 points` of macro F1 compared with fine-tuning only stages 3 and 4).
2. **Effect of smart cropping & minority oversampling:**
   - **Crop instead of shrink:** small catalogue objects with a large white border (`bbox_ratio < 0.75`) are kept as a native `224x224` cut-out at original resolution, while photos that already fill the frame stay unchanged.
   - **Strong gain on rare classes:** under-represented classes are drawn more often per epoch by the sampler and varied with the same augmentation, without augmentation artefacts leaking as a class shortcut. Validation loss decreases steadily over all 7 epochs to `1.0953`, without overfitting.
3. **New best score for text and image fusion (`05_fusion`):** Combined with the calibrated `LinearSVC` text model (`w = 0.60` text, `0.40` image), the fusion reaches **0.8640 weighted F1** (`+2.98 points` over text alone) and **0.8548 macro F1** (`+3.80 points` over text alone).